AD7031 Optimization Under Uncertainty

Homework 2 — Solutions

Due Oct 31, start of class

Problem 1. Mean and covariance of a random vector **(25 points)**

We have three independent identically distributed (i.i.d.) return data points:

	Tech	Energy	Bond
$\boldsymbol{\xi}^1$	0.00	0.04	0.03
$\boldsymbol{\xi}^2$	0.06	0.10	0.03
$\boldsymbol{\xi}^3$	0.12	0.07	0.03

Note that we use $\widehat{\mathbb{E}}[\cdot]$, $\widehat{\mathrm{Var}}[\cdot]$ and $\widehat{\mathrm{Cov}}[\cdot]$ for empirical estimates, to distinguish them from their true counterparts $\mathbb{E}[\cdot]$, $\mathrm{Var}[\cdot]$ and $\mathrm{Cov}[\cdot]$.

(a) **(8 pts) By hand**, compute the empirical mean $\widehat{\boldsymbol{\mu}} = \widehat{\mathbb{E}}[\tilde{\boldsymbol{\xi}}]$ and the empirical covariance matrix $\widehat{\Sigma} = \widehat{\mathrm{Cov}}[\tilde{\boldsymbol{\xi}}]$, and state the dimensions of $\widehat{\boldsymbol{\mu}}$ and $\widehat{\Sigma}$.

Solution. $\widehat{\mathbb{E}}$ averages over the three data points:

$$\widehat{\boldsymbol{\mu}} = \tfrac{1}{3}(\boldsymbol{\xi}^1 + \boldsymbol{\xi}^2 + \boldsymbol{\xi}^3) = \begin{pmatrix} 0.06 \\ 0.07 \\ 0.03 \end{pmatrix} \in \mathbb{R}^3.$$

The deviations $\boldsymbol{\xi}^s - \widehat{\boldsymbol{\mu}}$ are $(-0.06,\ -0.03,\ 0)$, $(0,\ 0.03,\ 0)$ and $(0.06,\ 0,\ 0)$, so $\widehat{\Sigma} = \frac{1}{3}\sum_s (\boldsymbol{\xi}^s - \widehat{\boldsymbol{\mu}})(\boldsymbol{\xi}^s - \widehat{\boldsymbol{\mu}})^\top$ is

$$\widehat{\Sigma} = \begin{pmatrix} 0.0024 & 0.0006 & 0 \\ 0.0006 & 0.0006 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3},$$

e.g. $\widehat{\Sigma}_{11} = \frac{1}{3}(0.06^2 + 0 + 0.06^2) = 0.0024$ and $\widehat{\Sigma}_{12} = \frac{1}{3}\big((-0.06)(-0.03) + 0 + 0\big) = 0.0006$. The Bond never moves, so its row and column are zero. (Dividing by $3 - 1 = 2$ instead, the unbiased sample covariance, is also accepted.)

(b) **(5 pts)** We used only three data points. In general, with N i.i.d. samples of $\tilde{\boldsymbol{\xi}}$, the empirical mean and covariance matrix can be written $\widehat{\boldsymbol{\mu}}_N$ and $\widehat{\Sigma}_N$. Consider the claim

$$\lim_{N\to\infty} \widehat{\boldsymbol{\mu}}_N = \mathbb{E}[\tilde{\boldsymbol{\xi}}] = \boldsymbol{\mu}, \qquad \lim_{N\to\infty} \widehat{\Sigma}_N = \mathrm{Cov}[\tilde{\boldsymbol{\xi}}] = \Sigma.$$

What does it mean? Explain in plain Korean. Is it true? If so, why?

Solution. 데이터를 많이 모을수록, 데이터로 계산한 평균과 공분산이 진짜 평균과 공분산에 점점 가까워진다는 뜻이다. 데이터가 무한히 많으면 둘은 같아진다.

It is true, with probability one, as long as $\boldsymbol{\mu}$ and Σ are finite. The reason is the **law of large numbers**: the average of i.i.d. samples converges to their expectation.

- $\widehat{\boldsymbol{\mu}}_N = \frac{1}{N}\sum_{s=1}^{N} \boldsymbol{\xi}^s \to \mathbb{E}[\tilde{\boldsymbol{\xi}}] = \boldsymbol{\mu}$.
- $\widehat{\Sigma}_N = \frac{1}{N}\sum_{s=1}^{N} \boldsymbol{\xi}^s \boldsymbol{\xi}^{s\top} - \widehat{\boldsymbol{\mu}}_N \widehat{\boldsymbol{\mu}}_N^\top$. The law of large numbers applied to $\tilde{\boldsymbol{\xi}}\tilde{\boldsymbol{\xi}}^\top$ gives $\frac{1}{N}\sum_s \boldsymbol{\xi}^s \boldsymbol{\xi}^{s\top} \to \mathbb{E}[\tilde{\boldsymbol{\xi}}\tilde{\boldsymbol{\xi}}^\top]$, and $\widehat{\boldsymbol{\mu}}_N \widehat{\boldsymbol{\mu}}_N^\top \to \boldsymbol{\mu}\boldsymbol{\mu}^\top$, so $\widehat{\Sigma}_N \to \mathbb{E}[\tilde{\boldsymbol{\xi}}\tilde{\boldsymbol{\xi}}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top = \Sigma$.

(c) **(7 pts)** Suppose your portfolio is $\bar{\boldsymbol{x}} = (\bar{x}_1, \bar{x}_2, \bar{x}_3)^\top$. From the table, compute, in terms of $\bar{\boldsymbol{x}}$,

- the empirical mean $\widehat{\mathbb{E}}[\tilde{\boldsymbol{\xi}}^\top \bar{\boldsymbol{x}}]$;
- the empirical variance $\widehat{\mathrm{Var}}[\tilde{\boldsymbol{\xi}}^\top \bar{\boldsymbol{x}}]$.

Solution. Write x_j for $\bar{x}_j$. The portfolio return for each data point is

$$\boldsymbol{\xi}^{1\top}\boldsymbol{x} = 0.04x_2 + 0.03x_3, \qquad \boldsymbol{\xi}^{2\top}\boldsymbol{x} = 0.06x_1 + 0.10x_2 + 0.03x_3, \qquad \boldsymbol{\xi}^{3\top}\boldsymbol{x} = 0.12x_1 + 0.07x_2 + 0.03x_3.$$

Averaging the three,

$$\widehat{\mathbb{E}}[\tilde{\boldsymbol{\xi}}^\top \boldsymbol{x}] = 0.06x_1 + 0.07x_2 + 0.03x_3.$$

The deviations from the mean are $-0.06x_1 - 0.03x_2$, $0.03x_2$ and $0.06x_1$, so

$$\widehat{\mathrm{Var}}[\tilde{\boldsymbol{\xi}}^\top \boldsymbol{x}] = \tfrac{1}{3}\left[(0.06x_1 + 0.03x_2)^2 + (0.03x_2)^2 + (0.06x_1)^2\right] = 0.0024x_1^2 + 0.0012x_1x_2 + 0.0006x_2^2.$$

x_3 drops out of the variance: the Bond adds no risk.

(d) **(5 pts)** Show that

- $\widehat{\mathbb{E}}[\tilde{\boldsymbol{\xi}}^\top \bar{\boldsymbol{x}}] = \widehat{\boldsymbol{\mu}}^\top \bar{\boldsymbol{x}}$;
- $\widehat{\mathrm{Var}}[\tilde{\boldsymbol{\xi}}^\top \bar{\boldsymbol{x}}] = \bar{\boldsymbol{x}}^\top \widehat{\Sigma}\, \bar{\boldsymbol{x}}$.

Solution. Write $\boldsymbol{x}$ for $\bar{\boldsymbol{x}}$. $\widehat{\mathbb{E}}$ is an average, so it is linear:

$$\widehat{\mathbb{E}}[\tilde{\boldsymbol{\xi}}^\top \boldsymbol{x}] = \widehat{\mathbb{E}}\Big[\sum_j \tilde{\xi}_j x_j\Big] = \sum_j x_j\, \widehat{\mathbb{E}}[\tilde{\xi}_j] = \widehat{\boldsymbol{\mu}}^\top \boldsymbol{x}.$$

Since $\tilde{\boldsymbol{\xi}}^\top \boldsymbol{x} - \widehat{\boldsymbol{\mu}}^\top \boldsymbol{x} = (\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})^\top \boldsymbol{x}$ is a number, its square is $\boldsymbol{x}^\top(\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})(\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})^\top \boldsymbol{x}$, so

$$\widehat{\mathrm{Var}}[\tilde{\boldsymbol{\xi}}^\top \boldsymbol{x}] = \widehat{\mathbb{E}}\big[\boldsymbol{x}^\top(\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})(\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})^\top \boldsymbol{x}\big] = \boldsymbol{x}^\top\, \widehat{\mathbb{E}}\big[(\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})(\tilde{\boldsymbol{\xi}} - \widehat{\boldsymbol{\mu}})^\top\big]\, \boldsymbol{x} = \boldsymbol{x}^\top \widehat{\Sigma}\, \boldsymbol{x}.$$

The same steps hold with $\mathbb{E}$, $\boldsymbol{\mu}$, Σ for any distribution. Check: with $\widehat{\Sigma}$ from (a), $\boldsymbol{x}^\top \widehat{\Sigma} \boldsymbol{x} = 0.0024x_1^2 + 2(0.0006)x_1x_2 + 0.0006x_2^2$, the same as (c).

Problem 2. Linear program duality **(25 points)**

Consider the following primal linear program and its dual, in the canonical form of the lecture note:

$$\text{(P)}\quad \begin{array}{ll} \min\limits_{\boldsymbol{x}\in\mathbb{R}^n} & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} & A\boldsymbol{x} \le \boldsymbol{b} \end{array} \qquad \text{(D)}\quad \begin{array}{ll} \max\limits_{\boldsymbol{\theta}\in\mathbb{R}^m} & -\boldsymbol{b}^\top \boldsymbol{\theta} \\ \text{s.t.} & \boldsymbol{c} + A^\top \boldsymbol{\theta} = \boldsymbol{0} \\ & \boldsymbol{\theta} \ge \boldsymbol{0} \end{array}$$

(a) **(8 pts)** Derive the dual problem (D) from the primal problem (P). **Show all the steps clearly.**

Solution. Attach $\boldsymbol{\theta} \ge \boldsymbol{0}$ to $A\boldsymbol{x} \le \boldsymbol{b}$:

$$L(\boldsymbol{x}, \boldsymbol{\theta}) = \boldsymbol{c}^\top \boldsymbol{x} + \boldsymbol{\theta}^\top (A\boldsymbol{x} - \boldsymbol{b}).$$

1. For fixed $\boldsymbol{x}$, $\max_{\boldsymbol{\theta}\ge\boldsymbol{0}} \boldsymbol{\theta}^\top(A\boldsymbol{x} - \boldsymbol{b})$ is 0 if $A\boldsymbol{x} \le \boldsymbol{b}$ (take $\boldsymbol{\theta} = \boldsymbol{0}$) and $+\infty$ otherwise (let $\theta_i \to \infty$ on a violated row). So
$$\text{(P)} = \min_{\boldsymbol{x}} \max_{\boldsymbol{\theta}\ge\boldsymbol{0}} L(\boldsymbol{x}, \boldsymbol{\theta}).$$
2. Swapping the operators can only lower the value (weak duality):
$$\min_{\boldsymbol{x}} \max_{\boldsymbol{\theta}\ge\boldsymbol{0}} L(\boldsymbol{x}, \boldsymbol{\theta}) \;\ge\; \max_{\boldsymbol{\theta}\ge\boldsymbol{0}} \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\theta}).$$
3. For fixed $\boldsymbol{\theta}$, $L = (\boldsymbol{c} + A^\top \boldsymbol{\theta})^\top \boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{\theta}$. Its minimum over $\boldsymbol{x} \in \mathbb{R}^n$ is $-\boldsymbol{b}^\top \boldsymbol{\theta}$ if $\boldsymbol{c} + A^\top \boldsymbol{\theta} = \boldsymbol{0}$, and $-\infty$ otherwise.
4. The outer max never picks a $\boldsymbol{\theta}$ with value $-\infty$, so $\boldsymbol{c} + A^\top \boldsymbol{\theta} = \boldsymbol{0}$ becomes a constraint. This is (D):
$$\begin{array}{ll} \max\limits_{\boldsymbol{\theta}} & -\boldsymbol{b}^\top \boldsymbol{\theta} \\ \text{s.t.} & \boldsymbol{c} + A^\top \boldsymbol{\theta} = \boldsymbol{0} \\ & \boldsymbol{\theta} \ge \boldsymbol{0}. \end{array}$$

(b) **(7 pts)** Consider the primal problem

$$\begin{array}{ll} \min\limits_{\boldsymbol{x}\in\mathbb{R}^2} & -4x_1 - 5x_2 \\ \text{s.t.} & 3x_1 + 2x_2 \le 60 \\ & x_1 + 3x_2 \le 48 \\ & x_1 \ge 3 \\ & x_1,\ x_2 \ge 0. \end{array}$$

- Give $\boldsymbol{c}$, A, $\boldsymbol{b}$.

- Write the dual problem (D) numerically.

Solution. $x_1 \geq 3$ becomes $-x_1 \leq -3$ and $x_2 \geq 0$ becomes $-x_2 \leq 0$; $x_1 \geq 0$ is implied by $x_1 \geq 3$ and dropped:

$$\boldsymbol{c} = \begin{pmatrix} -4 \\ -5 \end{pmatrix}, \qquad A = \begin{pmatrix} 3 & 2 \\ 1 & 3 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 60 \\ 48 \\ -3 \\ 0 \end{pmatrix}.$$

With $\boldsymbol{\theta} \in \mathbb{R}^4$, $-\boldsymbol{b}^\top \boldsymbol{\theta} = -60\theta_1 - 48\theta_2 + 3\theta_3$ and $\boldsymbol{c} + A^\top \boldsymbol{\theta} = \mathbf{0}$ gives two equations:

$$\begin{aligned} \max_{\boldsymbol{\theta} \in \mathbb{R}^4} \quad & -60\theta_1 - 48\theta_2 + 3\theta_3 \\ \text{s.t.} \quad & 3\theta_1 + \theta_2 - \theta_3 = 4 \\ & 2\theta_1 + 3\theta_2 - \theta_4 = 5 \\ & \theta_1, \theta_2, \theta_3, \theta_4 \geq 0. \end{aligned}$$

(Keeping the redundant row $-x_1 \leq 0$ adds a $\theta_5 \geq 0$ with $-\theta_5$ in the first equation; also correct.)

(c) **(7 pts)** Derive the primal problem (P) from the dual problem (D). **Show all the steps clearly.**

Solution. The same four steps, now for the max problem (D). Attach $\boldsymbol{x} \in \mathbb{R}^n$ (free) to the equality $\boldsymbol{c} + A^\top \boldsymbol{\theta} = \mathbf{0}$ and $\boldsymbol{s} \geq \mathbf{0}$ to $\boldsymbol{\theta} \geq \mathbf{0}$:

$$L(\boldsymbol{\theta}, \boldsymbol{x}, \boldsymbol{s}) = -\boldsymbol{b}^\top \boldsymbol{\theta} + \boldsymbol{x}^\top (\boldsymbol{c} + A^\top \boldsymbol{\theta}) + \boldsymbol{s}^\top \boldsymbol{\theta}.$$

1. For fixed $\boldsymbol{\theta}$, $\min_{\boldsymbol{x},\, \boldsymbol{s} \geq \mathbf{0}} L$ is $-\boldsymbol{b}^\top \boldsymbol{\theta}$ if $\boldsymbol{c} + A^\top \boldsymbol{\theta} = \mathbf{0}$ and $\boldsymbol{\theta} \geq \mathbf{0}$, and $-\infty$ otherwise. So
$$\text{(D)} = \max_{\boldsymbol{\theta}} \min_{\boldsymbol{x},\, \boldsymbol{s} \geq \mathbf{0}} L(\boldsymbol{\theta}, \boldsymbol{x}, \boldsymbol{s}).$$
2. Weak duality:
$$\max_{\boldsymbol{\theta}} \min_{\boldsymbol{x},\, \boldsymbol{s} \geq \mathbf{0}} L(\boldsymbol{\theta}, \boldsymbol{x}, \boldsymbol{s}) \;\leq\; \min_{\boldsymbol{x},\, \boldsymbol{s} \geq \mathbf{0}} \max_{\boldsymbol{\theta}} L(\boldsymbol{\theta}, \boldsymbol{x}, \boldsymbol{s}).$$
3. For fixed $(\boldsymbol{x}, \boldsymbol{s})$, $L = \boldsymbol{c}^\top \boldsymbol{x} + (A\boldsymbol{x} + \boldsymbol{s} - \boldsymbol{b})^\top \boldsymbol{\theta}$. Its maximum over $\boldsymbol{\theta} \in \mathbb{R}^m$ is $\boldsymbol{c}^\top \boldsymbol{x}$ if $A\boldsymbol{x} + \boldsymbol{s} = \boldsymbol{b}$, and $+\infty$ otherwise.
4. What survives is
$$\begin{aligned} \min_{\boldsymbol{x},\, \boldsymbol{s}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & A\boldsymbol{x} + \boldsymbol{s} = \boldsymbol{b} \\ & \boldsymbol{s} \geq \mathbf{0}. \end{aligned}$$
Here $\boldsymbol{s}$ is just the slack, so this is (P):
$$\begin{aligned} \min_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & A\boldsymbol{x} \leq \boldsymbol{b}. \end{aligned}$$

(d) **(3 pts)** Let $\boldsymbol{x}^\star$ and $\boldsymbol{\theta}^\star$ be optimal solutions of the primal problem (P) and the dual problem (D), respectively. What can you say about $\boldsymbol{c}^\top \boldsymbol{x}^\star$ and $\boldsymbol{b}^\top \boldsymbol{\theta}^\star$?

Solution. For any feasible $\boldsymbol{x}$ and $\boldsymbol{\theta}$, $\boldsymbol{c}^\top \boldsymbol{x} = -\boldsymbol{\theta}^\top A\boldsymbol{x} \geq -\boldsymbol{\theta}^\top \boldsymbol{b}$ (weak duality). For linear programs strong duality holds, so at the optimum
$$\boldsymbol{c}^\top \boldsymbol{x}^\star = -\boldsymbol{b}^\top \boldsymbol{\theta}^\star.$$
In (b): $\boldsymbol{x}^\star = (12, 12)$ and $\boldsymbol{\theta}^\star = (1, 1, 0, 0)$, and both values are -108.

Problem 3. Conic program duality **(25 points)**

Consider

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & \boldsymbol{x}^\top \Sigma \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{\mu}^\top \boldsymbol{x} \geq r \\ & \boldsymbol{e}^\top \boldsymbol{x} \leq 1 \\ & \boldsymbol{x} \geq \mathbf{0} \end{aligned}$$

where $\boldsymbol{e} = [1, 1, \ldots, 1]^\top$ is the all-ones vector and r is the target return.

(a) **(12 pts)** Recall the second-order cone

$$\mathrm{SOC}_Q^{n+1} = \big\{(\boldsymbol{x}, t) \in \mathbb{R}^n \times \mathbb{R} \;:\; \boldsymbol{x}^\top Q\, \boldsymbol{x} \le t^2,\; t \ge 0\big\}, \qquad Q \succ 0.$$

Write the model in the conic form

$$\begin{aligned} \min \quad & \boldsymbol{c}^\top \boldsymbol{z} \\ \text{s.t.} \quad & A\boldsymbol{z} \le \boldsymbol{b} \\ & \boldsymbol{z} \in C. \end{aligned}$$

- Give $\boldsymbol{z}$ (the decision variable), $\boldsymbol{c}$, A, $\boldsymbol{b}$.
- Write C.

Solution. Minimizing $\boldsymbol{x}^\top \Sigma \boldsymbol{x}$ and minimizing $\sqrt{\boldsymbol{x}^\top \Sigma \boldsymbol{x}}$ give the same $\boldsymbol{x}$. The epigraph reformulation with a new variable t is

$$\begin{aligned} \min_{\boldsymbol{x},\, t} \quad & t \\ \text{s.t.} \quad & \boldsymbol{x}^\top \Sigma \boldsymbol{x} \le t^2,\; t \ge 0 \\ & \boldsymbol{\mu}^\top \boldsymbol{x} \ge r \\ & \boldsymbol{e}^\top \boldsymbol{x} \le 1 \\ & \boldsymbol{x} \ge \boldsymbol{0}. \end{aligned}$$

With $\boldsymbol{z} = [\boldsymbol{x},\; t]^\top \in \mathbb{R}^{n+1}$:

$$\boldsymbol{c} = \begin{pmatrix} \boldsymbol{0} \\ 1 \end{pmatrix}, \qquad A = \begin{pmatrix} -\boldsymbol{\mu}^\top & 0 \\ \boldsymbol{e}^\top & 0 \\ -I_n & \boldsymbol{0} \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} -r \\ 1 \\ \boldsymbol{0} \end{pmatrix} \quad \begin{matrix} \text{return} \\ \text{budget} \\ \boldsymbol{x} \ge \boldsymbol{0} \end{matrix}$$

with $\boldsymbol{c} \in \mathbb{R}^{n+1}$, $A \in \mathbb{R}^{(n+2)\times(n+1)}$ and $\boldsymbol{b} \in \mathbb{R}^{n+2}$. The cone is

$$C = \mathrm{SOC}_\Sigma^{n+1} = \big\{(\boldsymbol{x}, t) \in \mathbb{R}^n \times \mathbb{R} \;:\; \boldsymbol{x}^\top \Sigma\, \boldsymbol{x} \le t^2,\; t \ge 0\big\}.$$

Check: $\boldsymbol{c}^\top \boldsymbol{z} = t$, and $A\boldsymbol{z} \le \boldsymbol{b}$ reads $\boldsymbol{\mu}^\top \boldsymbol{x} \ge r$, $\boldsymbol{e}^\top \boldsymbol{x} \le 1$, $\boldsymbol{x} \ge \boldsymbol{0}$.

(b) **(13 pts)** With $\Sigma = I$, the dual problem of the second-order cone program in (a) is

$$\begin{aligned} \max \quad & r\,\theta_1 - \theta_2 \\ \text{s.t.} \quad & \theta_1,\; \theta_2 \ge 0, \quad \boldsymbol{\theta}_3 \in \mathbb{R}^n_+ \\ & \|\theta_2 \boldsymbol{e} - \theta_1 \boldsymbol{\mu} - \boldsymbol{\theta}_3\|_2 \;\le\; 1. \end{aligned}$$

Derive it. **Show all the steps clearly.**

Solution. In (a), we have

$$C = \mathrm{SOC}_\Sigma^{n+1} = \big\{(\boldsymbol{x}, t) \in \mathbb{R}^n \times \mathbb{R} \;:\; \boldsymbol{x}^\top \Sigma\, \boldsymbol{x} \le t^2,\; t \ge 0\big\}.$$

With $\Sigma = I$, our cone C becomes

$$C = \mathrm{SOC}_I^{n+1} = \big\{(\boldsymbol{x}, t) \in \mathbb{R}^n \times \mathbb{R} \;:\; \boldsymbol{x}^\top I\, \boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{x} = \|\boldsymbol{x}\|_2^2 \le t^2,\; t \ge 0\big\}.$$

The primal problem from (a), with a dual variable next to each constraint in $A\boldsymbol{z} \le \boldsymbol{b}$, is

$$\begin{aligned} \min \quad & \boldsymbol{c}^\top \boldsymbol{z} \\ \text{s.t.} \quad & A\boldsymbol{z} \le \boldsymbol{b} \\ & \boldsymbol{z} \in C \end{aligned} \quad = \quad \begin{aligned} \min_{\boldsymbol{x},\, t} \quad & t \\ \text{s.t.} \quad & \boldsymbol{\mu}^\top \boldsymbol{x} \ge r \qquad (:\, \theta_1 \in \mathbb{R}_+) \\ & \boldsymbol{e}^\top \boldsymbol{x} \le 1 \qquad (:\, \theta_2 \in \mathbb{R}_+) \\ & \boldsymbol{x} \ge \boldsymbol{0} \qquad (:\, \boldsymbol{\theta}_3 \in \mathbb{R}^n_+) \\ & (\boldsymbol{x}, t) \in C. \end{aligned}$$

Together, $\boldsymbol{\theta} = (\theta_1, \theta_2, \boldsymbol{\theta}_3) \in \mathbb{R}^{n+2}_+$. The steps of Problem 2(a), keeping $\boldsymbol{z} \in C$ as a constraint:

$$L(\boldsymbol{z}, \boldsymbol{\theta}) = \boldsymbol{c}^\top \boldsymbol{z} + \boldsymbol{\theta}^\top (A\boldsymbol{z} - \boldsymbol{b}).$$

1. The primal problem is written as

$$\min_{\boldsymbol{z} \in C} \max_{\boldsymbol{\theta} \in \mathbb{R}^{n+2}_+} L(\boldsymbol{z}, \boldsymbol{\theta}).$$

Switching min and max, the dual problem is

$$\max_{\boldsymbol{\theta} \in \mathbb{R}^{n+2}_+} \min_{\boldsymbol{z} \in C} L(\boldsymbol{z}, \boldsymbol{\theta}).$$

2. Collecting the terms in $\boldsymbol{z}$,

$$\max_{\boldsymbol{\theta}\in\mathbb{R}^{n+2}_+} \min_{\boldsymbol{z}\in C} L(\boldsymbol{z},\boldsymbol{\theta}) = \max_{\boldsymbol{\theta}\in\mathbb{R}^{n+2}_+} -\boldsymbol{b}^\top\boldsymbol{\theta} + \min_{\boldsymbol{z}\in C} (\boldsymbol{c}+A^\top\boldsymbol{\theta})^\top\boldsymbol{z},$$

where

$$\min_{\boldsymbol{z}\in C} (\boldsymbol{c}+A^\top\boldsymbol{\theta})^\top\boldsymbol{z} = \begin{cases} 0 & \text{if } \boldsymbol{c}+A^\top\boldsymbol{\theta}\in C^* \\ -\infty & \text{otherwise.} \end{cases}$$

With $\boldsymbol{y} = \boldsymbol{c}+A^\top\boldsymbol{\theta}$:

- $\boldsymbol{y}\in C^*$: $\boldsymbol{y}^\top\boldsymbol{z}\geq 0$ for all $\boldsymbol{z}\in C$, and $\boldsymbol{z}=\boldsymbol{0}\in C$, so the min is 0.
- $\boldsymbol{y}\notin C^*$: there is $\bar{\boldsymbol{z}}\in C$ with $\boldsymbol{y}^\top\bar{\boldsymbol{z}}<0$, and $\lambda\bar{\boldsymbol{z}}\in C$ for all $\lambda\geq 0$ (cone), so $\lambda\,\boldsymbol{y}^\top\bar{\boldsymbol{z}}\to-\infty$ as $\lambda\to\infty$.

Therefore, the dual problem is

$$\begin{aligned} \max \quad & -\boldsymbol{b}^\top\boldsymbol{\theta} \\ \text{s.t.} \quad & \boldsymbol{\theta}\in\mathbb{R}^{n+2}_+ \\ & \boldsymbol{c}+A^\top\boldsymbol{\theta}\in C^*. \end{aligned}$$

Substituting $\boldsymbol{c}$, A, $\boldsymbol{b}$ from (a),

$$-\boldsymbol{b}^\top\boldsymbol{\theta} = -\begin{pmatrix}-r & 1 & \boldsymbol{0}^\top\end{pmatrix}\begin{pmatrix}\theta_1\\ \theta_2\\ \boldsymbol{\theta}_3\end{pmatrix} = r\theta_1-\theta_2,$$

$$\boldsymbol{c}+A^\top\boldsymbol{\theta} = \begin{pmatrix}\boldsymbol{0}\\ 1\end{pmatrix} + \begin{pmatrix}-\boldsymbol{\mu} & \boldsymbol{e} & -I_n\\ 0 & 0 & \boldsymbol{0}^\top\end{pmatrix}\begin{pmatrix}\theta_1\\ \theta_2\\ \boldsymbol{\theta}_3\end{pmatrix} = \begin{pmatrix}\theta_2\boldsymbol{e}-\theta_1\boldsymbol{\mu}-\boldsymbol{\theta}_3\\ 1\end{pmatrix}.$$

With $\Sigma = I$, the second-order cone is self-dual (week 6), so

$$\boldsymbol{c}+A^\top\boldsymbol{\theta} = \begin{pmatrix}\theta_2\boldsymbol{e}-\theta_1\boldsymbol{\mu}-\boldsymbol{\theta}_3\\ 1\end{pmatrix} \in C^* = \left\{(\theta_2\boldsymbol{e}-\theta_1\boldsymbol{\mu}-\boldsymbol{\theta}_3,\ 1)\in\mathbb{R}^n\times\mathbb{R}\ :\ \|\theta_2\boldsymbol{e}-\theta_1\boldsymbol{\mu}-\boldsymbol{\theta}_3\|_2\leq 1\right\},$$

and the dual is

$$\begin{aligned} \max_{\theta_1,\,\theta_2,\,\boldsymbol{\theta}_3} \quad & r\,\theta_1-\theta_2 \\ \text{s.t.} \quad & \theta_1,\ \theta_2\geq 0, \quad \boldsymbol{\theta}_3\in\mathbb{R}^n_+ \\ & \|\theta_2\boldsymbol{e}-\theta_1\boldsymbol{\mu}-\boldsymbol{\theta}_3\|_2\ \leq\ 1. \end{aligned}$$